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DEVELOPMENT ARTICLE

The role of help-seeking from ChatGPT in digital game-based learning

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Abstract
This study explores the roles of students' help-seeking profiles when seeking help from AI chatbots, specifically ChatGPT, in a digital game-based learning environment, *Summon of Magicrystal*. The study involved 102 middle school students who played an online game with the provision of ChatGPT and sought help from ChatGPT while solving physics problems. The results revealed that students' help-seeking profiles, help-seeking threats, help-seeking avoidance, and instrumental help-seeking were positively correlated. Students' instrumental help-seeking profile has a positive effect on game performance/engagement, while students' avoidance help-seeking profile has a positive effect on the number of game attempts. The findings highlight the importance of students' help-seeking profiles when considering designing AI-assisted game-based learning environments to better support students' science learning.

Keywords ChatGPT · Help-seeking profiles · Digital game-based learning · Game performance

Introduction

Digital game-based learning (DGBL) has become a prevalent instructional approach to engage students in learning activities. DGBL not only enhances educational outcomes (Clark et al., 2015) but also offers a context where students can apply practical skills in learning environments. Along with the evolution of DGBL, the emerging Artificial Intelligence (AI) technologies have increasingly been integrated to provide more sophisticated and personalized learning experiences that cater to individual learners' needs (Brisson et al, 2021; Wu et al, 2023).

The primary goal of this study is to investigate the interplay between help-seeking behaviors and the use of AI technologies within a DGBL environment. Specifically, we

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aim to understand how distinct help-seeking profiles influence players' behavior and learning outcomes during gameplay. Help-seeking, often considered a self-regulated learning strategy, is defined as the process by which learners seek help when needed (Pintrich, 2000). It has been shown to enhance students' learning outcomes (Fong et al, 2023) and plays a critical role in enabling learners to overcome obstacles and attain their learning goals. Prior research also highlights that students' help-seeking behaviors can alter the success and optimal learning experience in DGBL (Xu et al., 2022).

Further, we consider the unique opportunities that AI technologies present in facilitating help-seeking within DGBL. First, DGBL provides a unique environment for help-seeking, as help sources and channels can provide scaffolds different from those in traditional classroom settings or online learning (Hong & Hwang, 2012; Yang et al, 2022). In the context of help-seeking, students in traditional classrooms predominantly seek informal help from teachers and peers (Karabenick & Knapp, 1991). In contrast, those in DGBL environments may rely more on in-game tutorials, hints, or automated feedback (Jimenez et al., 2014). AI-powered DGBL further expands help-seeking options by incorporating external resources such as chatbots and intelligent agents.

Therefore, the current study attempts to explore how help-seeking profiles could play and influence learners' help-seeking behavior in the gameplay. The findings of this study would not only extend current understandings of help-seeking in DGBL, but also shed light on the promotion of using AI for seeking help in DGBL.

Next, we succinctly review pertinent literature on help-seeking profiles and AI's impact on learning, and specific dynamics within DGBL, followed by proposed research questions and methodology carried out for investigations.

Literature review

Digital game-based learning (DGBL)

Prensky (2001) suggested digital educational games should include elements such as rules, goals, outcomes, feedbacks, conflict/challenge/competition, interaction, narratives/story, and adaptivity. Many other digital games also include technology and fun as crucial elements of digital educational games (Amory et al., 1999). DGBL can facilitate learning through various interactions, such as interactions among peers, interactions with avatars, and/or interactions with the learning environment (Ke, 2016; Young et al., 2012). The central premise of digital game-based learning (DGBL) is that it improves students' motivation, which, in turn, may enhance learning outcomes (e.g., Chen & Law, 2016; Clark, et al, 2015). In addition, different features of DGBL, such as competition, feedback, and adaptive challenges, may also provide practices and engagement opportunities that improve knowledge acquisition, knowledge retention, and problem-solving (Chen et al., 2019, 2020, 2023; Law & Chen, 2016).

Although digital game-based learning (DGBL) has demonstrated potential and some empirical evidence of success in educational environments, certain types of games and game features have yielded inconclusive results. For instance, Sitzmann (2011) found inconclusive evidence of simulation games where it was used as a standalone intervention. Barzilai and Blau (2014) found that scaffolding within DGBL might negatively impact students' perception of their learning experience, as it can disrupt the immersive flow of the game. Limitations of such DGBLs may be addressed by incorporating AI-help-seeking that provides students' agency with the scaffold of their choice. Another constant concern regarding game-based learning is that poorly designed scaffolding or gameplay mechanics can diminish the element of fun, reducing engagement. Flow theory suggests that students' perceived difficulties in the tasks can affect their flow state (Csikszentmihalyi, 1990). AI-help-seeking may provide the personalized support students need to maintain flow and enhance their enjoyment of the game.

Help-seeking

Help-seeking may take place at all five phases of the Nelson-Le Gall (1985) model, recognizing the need for assistance, opting to seek help, identifying potential sources of assistance, employing strategies to acquire help, and assessing the assistance received. Throughout these phases, students' profiles and goals toward help-seeking may explain the reasons behind their inclination to seek help and their preferences for specific sources of support. There are four help-seeking profiles: instrumental help-seeking, help-seeking avoidance, help-seeking threats, and formal/informal help-seeking (Karabenick, 2003). Instrumental help-seeking, an example of adaptive help-seeking, has been explored in various domains and learning contexts. Students with instrumental help-seeking tendency prioritize identifying sources of aid while working through problems independently until solutions are reached (Karabenick, 2003). Instead of just looking for an answer, learners try to get help in solving academic problems by asking for hints or directions (Karabenick, 2003). Instrumental help-seeking has been found to be an active strategy and positively influences academic learning outcomes (Karabenick & Newman, 2009). On the other hand, help-seeking avoidance refers to the behaviors where students need assistance but decide not to proactively seek it (Ryan et al., 1998). Help-seeking avoidance is often associated with low self-efficacy and learning outcomes (Ryan & Shim, 2012). The academic performance of students with a help-seeking avoidance profile was found to be weaker, and this trait displayed a negative correlation with instrumental help-seeking (Karabenick, 2003). The third profile is help-seeking threat, where learners may prevent seeking help to protect their self-esteem (Karabenick, 2003; Karabenick & Knapp, 1991). Help-seeking threat has been found positively correlated with help-seeking avoidance (Karabenick, 2003). The last help-seeking profile, formal/informal help-seeking, focuses on the source of help-seeking (Karabenick, 2011). When learners decide to seek assistance, they may select various sources depending on their preferences. Certain individuals might opt to request assistance from their instructors (formal), while others may turn to their peers or utilize technology (informal) for help. There have been inconsistent results regarding the impact of different sources of help on academic performance.

Studies on help-seeking have been conducted both in traditional classroom settings and online learning such as MOOC, Blackboard, or Forum (Aleven et al., 2003; Kizilcec et al., 2017; Koc & Liu, 2016; Mahasneh et al., 2012). The effect of help-seeking profiles on learning outcomes can be significant in online learning environments (Huang et al., 2024). With the advancement of modern technologies, the digital game-based learning (DGBL) environment has evolved significantly to provide more immersive, interactive, and engaging learning experiences. The multitude of assistance options provided by DGBL is different from traditional classroom or online settings. For example, besides formal help-seeking from teachers or informal help-seeking from peers, students have the option to seek help from technology such as AI chatbots. Because students are not required to seek help from their peers or in front of their peers when an AI chatbot is provided, their help-seeking tendencies may not be the same as they may not need to avoid help-seeking or feel threats in an AI-assisted DGBL environment.

AI-assisted DGBL and help-seeking

We are in an era characterized by dynamic technological shifts, and artificial intelligence (AI) is a key driver of this transformation. A prominent impact of AI is through AI-driven natural language processing models that enable machines to comprehend, interpret, and generate human language. This is exemplified by applications such as chatbots. In education, leveraging AI technologies that mimic human intelligence can not only deliver customized guidance, support, and feedback to students, but also aid teachers and policymakers in decision-making processes (Hwang et al., 2020).

DGBL has emerged as a promising approach in education, as younger generations are already engaged in computer and mobile games from an early age. While DGBL aims to achieve specific educational objectives, it also offers immersive experiences for knowledge acquisition and skill development (Hawlitschek & Joeckel, 2017; Parasleva et al., 2010; Wouters et al., 2013). The use of AI in DGBL holds many promising benefits that can enhance learning experience for students. AI can stimulate human behavior, optimize game strategies, and enhance player engagement (Dyulicheva & Glazieva, 2021). Notably, AI, such as virtual or intelligent characters, can tailor game's content and challenges to match individual learners' needs (Pacella & López-Pérez, 2018; Tumenayu et al., 2014). AI can also be incorporated in collaborative learning environments to facilitate problem solving (Lee et al, 2021). A recent popular AI chatbot tool, ChatGPT is gaining significant attention due to its capacity to swiftly and accurately process substantial volumes of data (Haleem et al., 2022). ChatGPT has the potential to provide personalized assistance by explanations, clarifications, and answering questions in real-time which can induce transformative impacts in teaching and learning (Dwivedi et al., 2023). ChatGPT can provide assistance whenever a student needs it, even outside of regular school hours, where teachers and peers have limited availability. ChatGPT can adapt to individual learning preferences and paces, providing tailored explanations and examples that suit each student's needs (Nikolic et al., 2023). In addition, help-seekers' anonymity can be a crucial factor in understanding help-seeking behavior, because students are likely to abstain from seeking help when their identities are discernible (Nadler & Porat, 1978). ChatGPT offers a high degree of anonymity, allowing students to ask questions without fear of being perceived as incompetent or uninformed. To extend the pedagogical implications of how ChatGPT can be used in education, individuals' help-seeking profiles may play an antecedent role in choosing and using ChatGPT for assistance during the learning process. Lo (2023) also calls for studies empirically evaluating the effects of using ChatGPT on learning outcomes.

We found that learner's help-seeking profiles are related to their online engagement behaviors in an online class (Huang et al., 2024). However, there has been no investigation into the relationship between learners' help-seeking profiles and behaviors in the AI-assisted DGBL context. Decades of research on computer-assisted instruction provide valuable insights into how learners' help-seeking behaviors can be effectively supported in digital environments. In the current study, we attempt to gain a more comprehensive understanding of learners' help-seeking behaviors by using AI-assisted DGBL that provides a traceable record of the learning process in order to shed light on the learning behaviors and strategies of learners.

The present study and research questions

In the aforementioned literature reviews, a few research gaps have been identified. First, the majority of research on help-seeking has been dedicated to exploring the connection between help-seeking profiles and learning outcomes in typical classroom settings or online learning environments (e.g., Fong et al., 2023). Game-based learning environments provide a unique opportunity for students to learn, and help-seeking profiles may affect students' learning differently in such environments. Specifically, students' help-seeking profiles could have varying effects on both their behaviors during gameplay and the outcomes of their learning. Second, generative AI is still a relatively new phenomenon in the area of education. Generative AI can be an effective tool to assist students in learning. However, the dynamics of how students' help-seeking behaviors will unfold in a generative AI environment remain uncertain. To bridge these gaps, this study implemented an AI-assisted game-based learning and aimed to investigate the following research questions and hypotheses:

(1) How do different help-seeking profiles relate to each other in a ChatGPT-embedded GBL environment?

Hypothesis 1 There is a significant relationship between at least two help-seeking profiles (help-seeking threats, help-seeking avoidance, instrumental help-seeking, formal/informal help-seeking) in a ChatGPT-embedded GBL environment.

(2) How do help-seeking profiles relate to students' gameplay behaviors (gameplay scores, gameplay time, and gameplay attempts)?

Hypothesis 2 There is a significant relationship between help-seeking profiles (help-seeking threats, help-seeking avoidance, instrumental help-seeking, formal/informal help-seeking) and students' gameplay behaviors (highest game score, game playing time, and attempts) in a game-based learning environment.

(3) How do help-seeking profiles relate to help-seeking behaviors?

Hypothesis 3 There is a significant relationship between help-seeking profiles (help-seeking threats, help-seeking avoidance, instrumental help-seeking, formal/informal help-seeking) and help-seeking behaviors (Number of Questions posted in ChatGPT (a.k.a. GPT Questions) and time engaged in ChatGPT (a.k.a. GPT Time).

(4) How help-seeking profiles relate to learning outcomes?

Hypothesis 4 There is a significant relationship between different help-seeking profiles (help-seeking threats, help-seeking avoidance, instrumental help-seeking, formal/informal help-seeking) and learning outcomes.

Method

Participants

A total of 102 seventh-grade students (49 females and 53 males) were recruited from six classes in two different public middle schools in central Taiwan. The socioeconomic status of these students was largely homogeneous, with the majority of participants coming from similar backgrounds. A self-reported survey confirmed no statistically significant differences among the students in terms of their motivation to play educational games or their previous experience in using games to learn content knowledge ($p > 0.05$). Seventh-grade graders were chosen for this study because this age group of students is becoming increasingly influential in digital engagement, and their cognitive, social, and emotional development makes them particularly receptive to the educational impacts. Besides, this age group is sufficiently mature to provide reliable self-reported data. Additionally, the online game implemented in this study is primarily for middle school students, and the game's content is aligned with their curriculum.

Context of learning: Summon of Magicrystal

Summon of Magicrystal is a problem-based science inquiry learning game designed for secondary students to learn about the motion of objects. The game is played from a first-person perspective, allowing players to immerse themselves in the game environment. The game is controlled through a combination of mouse and keyboard inputs, with players using these controls to interact with objects, characters, and the game interface. *Summon of Magicrystal* is classified as a role-playing game (RPG) with adventure elements, as players assume the role of a character in a story-driven game world and engage in problem-solving activities. The game's design and development involved a multidisciplinary team that includes programming experts, science teachers, instructional designers, and educational researchers, ensuring a balance between technical excellence and pedagogical effectiveness. The game is built using JavaScript's superset, TypeScript because of its robust and scalable application design. The validity of the game as a learning tool has been established in our previous studies (Chen et al., 2019, 2023).

Fig. 1 Summon of magicrystal game interface

The learning objectives for the game are (1) to understand how these forces affect the movement and stability of objects during transportation and (2) to predict and analyze the effects of acceleration and deceleration on the soap's movement within the transport vehicle. As shown in Fig. 1, the game board includes game tasks, game rules, and the knowledge book. The game task involves players delivering soap bars at least seven times to complete the objective. In the game, soap serves as the primary object for players to manipulate, with its movement demonstrating key concepts such as friction, acceleration, and deceleration. Players must account for the effects of slight friction between the soap and the transport vehicle, as well as between the soap bars themselves, when planning their delivery strategies. These mechanics are designed to help players intuitively understand the physics concepts of friction, acceleration, and deceleration through experiential learning. The game rules provide clear guidelines on how the vehicle's acceleration and deceleration can be controlled. The knowledge book explains relevant science concepts such as friction, forces, motion, and acceleration. This game also incorporates a generative artificial intelligent chatbot tool to enhance the gaming experience. The chatbot is powered by the GPT version 3.5-turbo model and serves as a conversational AI agent to generate messages that enable students to engage in dynamic text-based interactions. GPT-3.5-turbo is chosen for its efficiency and cost-effectiveness, providing reasonable processing speeds and flexibility essential for digital game-based learning (DGBL). Students can click on the *AI-help* button on the right of the game interface and ask questions. In response, the chatbot offers immediate responses to students' inquiries.

As shown in Fig. 2, game elements involve game scores, feedback, progress tracking, and status reports. Game scores are awarded based on the completion of the tasks, accuracy, or speed. Feedback helps the players understand their performance and offers insights

Fig. 2 AI-assisted summon of magicrystal

into what actions were successful or where improvements can be made. The progress tracking includes comprehensive data about the total duration a player spends on a game, the number of tries a player takes to complete a level, numbers of soaps delivered successfully, and numbers of successful transportation. Status report element provides an overview of the player's achievements using a specialized three-star rating system to visually communicate key milestones.

Instruments

To measure students' help-seeking profiles, a 12-item questionnaire was adapted from Karabenick (2003) and Ryan and Shim (2012). The measurement included four subscales: help-seeking threat, avoidance of help-seeking, instrumental help-seeking goal, and formal versus informal help-seeking. The initial items regarding seeking assistance from this course were altered to reflect seeking assistance from AI. For example, the original item, "If I didn't understand something in this class, I would guess rather than ask someone for assistance." was revised to "If I didn't understand something in playing *Summon of Magicrystal*, I would guess rather than ask someone for assistance." And "I would ask someone to give me hints or clues rather than the answer." was revised to "When playing *Summon of Magicrystal,* I would ask AI to give me hints or clues rather than the answer." On a 5-point scale, spanning from "1-Not at all true of me" to "5-Very true of me" students provided their evaluations of agreement with each item. Karabenick (2003) documented that the subscales exhibited satisfactory levels of internal consistency. Items, descriptive statistics, and reliability coefficients for help-seeking profiles are presented in the Appendix. In addition to students' help-seeking profiles, game performance/engagement and ChatGPT behaviors were also retrieved from the system. Game performance/engagement included highest game score, gameplay time, and attempts (Chen & Law, 2016). The *highest game score* refers to the points the player gains based on their performance in the game. Game points or score is accumulated by successfully completing levels, with advanced levels with higher points reward compared to basic levels. *Gameplay time* refers to the amount of time a player spends in playing the game. *Attempts* refers to the number of times a player tries to complete the game. ChatGPT behaviors included the numbers of questions asked (a.k.a. *GPT Questions*) and the time spent on the ChatGPT forum (a.k.a. *GPT Time*). A set of sixteen multiple-choice questions was used to assess students' *learning outcomes* (posttest). The test items were adopted from Chen et al. (2019). The items were validated by two experienced science teachers to confirm their relevance to the game content. The reliability of the learning outcomes posttest is demonstrated by a Cronbach's //alpha of 0.80. An example of a multiple-choice question is "If it takes 5 s for a cell to divide once. How much time will it take to divide 15 times?". The associated answers were (A) Three seconds (B) Seventy-five seconds (C) One minute (D) Three minutes. Each correct answer was awarded one point, with no points deducted for incorrect answers. Test scores were calculated based on the percentage of correctly answered questions.

Study procedure

The study was conducted over a period of two weeks, with participants engaging in two sessions lasting 45 min each. It took place in computer labs at two different public middle schools in central Taiwan. From each school, three seventh-grade classes were randomly chosen to participate, and only students in these selected classes were included after informed consent was obtained. At the beginning of the study, participants completed a demographic survey which helped further confirm the homogeneity of the cohort in terms of age, socioeconomic status, and prior experience with digital games. Following this, the researchers provided a brief introduction to the study objectives and the game environment. Each student was given unique login credentials to access the game, and was informed about the integration of a ChatGPT-powered chatbot within the game, which was designed to facilitate interaction and support throughout the learning process. At the conclusion of the study, participants completed an online survey to assess their help-seeking profiles and a learning outcome test to measure the educational impacts.

Data analytical approaches

Three steps were executed to carry out the data analysis. The initial step consisted of determining the reliability of the scales. The scales displayed acceptable reliability, where Cronbach's α ranging from 0.72 to 0.96, except for Formal/Informal Help-Seeking, which had a Cronbach's α of 0.35 (Hair et al., 2006). After investigation, one survey item with reverse wording was found to be unreliable. As a result, we deleted the reverse wording survey item; the 2-item survey was reasonably reliable with Cronbach α of 0.65.

Second, we ensured the variables were normally distributed. Skewness and Kurtosis statistics were run and most of the variables except (1) attempts, which represents the number of attempts students played, (2) GPT Questions, which represents how many questions the students posted to ChatGPT, and (3) GPT Time, how long students engaged in the ChatGPT session. The distribution of attempts was highly skewed to the right. As a result, we

Table 1 Descriptive statistics of help-seeking profiles, game performance/engagement data and help-seeking behaviors

Variable	Mean	SD	Min	Max
Help-seeking profiles				
Help-seeking threats	2.36	1.08	1	5
Help-seeking avoidance	2.30	0.99	1	5
Instrumental help-seeking	3.25	1.12	1	5
Formal/informal help-seeking	2.89	1.01	1	5
Game performance/engagement				
Highest game score	183.10	161.12	0	536.00
Game playing time	1214.46	770.08	13.03	3581.40
Attempts	3.11	2.87	1	22
Learning outcomes (posttest)	49.26	12.94	18.75	87.50
Help-seeking behaviors (AI-Help)				
GPT questions	3.98	4.62	0.00	19.00
GPT time	205.93	234.58	0.00	1232.90

transformed the data by taking a natural log of the data such that skewness and kurtosis are smaller than 1.

Finally, we ran Pearson correlations among the help-seeking variables, the gameplay variables, posttest, and the help-seeking behaviors to test the four research questions: the relationships among the help-seeking profiles, the relationships between help-seeking profiles and gameplay behaviors, the relationships between help-seeking profiles and help-seeking behaviors, and the relationships between help-seeking profiles and learning outcomes.

Results

Table 1 presents the descriptive statistics of the variables. Our students' help-seeking profiles were all somewhat close to the middle of the 5-point scale (between 2.30 for help-seeking avoidance and 3.25 for instrumental help-seeking). On average, students spent 1214.46 s on gameplay. The average game score was 183.10, and the average learning outcome was 49.26. In addition, on average, they spent 205.93 s asking ChatGPT for four questions.

The correlations among the variables are presented in Table 2, which addresses the four research questions. Regarding RQ1, threats and avoidance were related positively, and they were not significantly related to instrumental help-seeking. Instrumental help-seeking was correlated with formal vs informal help-seeking. Regarding the relationships between help-seeking profiles and gameplay (RQ2), instrumental help-seeking is significantly related to highest game score, playtime and number of attempts. Avoidance help-seeking as well as formal vs informal help-seeking were positively related to the number of attempts. The more students prefer teachers/peers for a source of help compared to AI, the more they attempt the game. Regarding RQ 3 & 4, there were no significant correlations between help-seeking profiles and AI-help behaviors, nor between help-seeking profiles and students' posttest outcomes.

Table 2 Correlations among help-seeking profiles, AI-help, and game performance/engagement data

Variables	1	2	3	4	5	6	7	8	9	10
1. Help-seeking threats	1.00	0.63*	0.13	0.18	− 0.12	− 0.09	0.15	− 0.01	0.12	0.04
2. Help-seeking avoidance		1.00	0.17	0.13	− 0.10	− 0.05	0.20*	− 0.07	0.05	0.01
3. Instrumental help-seeking			1.00	0.42*	0.29*	0.21*	0.23*	0.10	− 0.04	0.11
4. Formal/informal help-seeking				1.00	0.17	0.14	0.23*	0.04	0.03	0.10
5. Highest game score					1.00	− 0.05	0.08	− 0.01	− 0.23*	− 0.15
6. Gameplay time						1.00	0.47*	− 0.06	0.13	0.18
7. Attempts							1.00	− 0.07	− 0.00	0.09
8. Learning outcomes								1.00	0.14	0.13
9. GPT questions									1.00	0.69*
10. GPT time										1.00

Attempts, GPT Questions, and GPT Time were taken a natural log for the correlation analysis

Discussion and conclusion

Before discussing the relationships among the help-seeking profiles, we observed that the help-seeking profiles could be a little bit different from those in similar studies (e.g., Huang & Law, 2022; Pajares et al., 2004; Ryan & Pintrich, 1997). Even though in general, students' threat and avoidance help-seeking tendencies were lower than the middle point of the scale as the literature indicated, the mean of threat and avoidance help-seeking were 2.36 and 2.30 on a 5-point scale. In other words, they were not very reluctant to seek help compared to the students in other studies where the mean threat and avoidance scores 2.25 on an 8-point scale (Pajares et al., 2004) or 1.44 on a 5-point scale (Huang & Law, 2022). Their instrumental help-seeking was also stronger. These tendencies might be due to the fact that they were in a DGBL environment, as students might not think too much about learning, but focus on playing the games. This finding uniquely shows the potential of incorporating generative AI in a game-based learning environment, especially to improve students' help-seeking during gameplay as many game players are used to seeking help in commercial games.

The first research question examined the relationship among the help-seeking profiles. Our results are somewhat consistent with the help-seeking literature. First, threats and avoidance are positively related as expected. Students who viewed help-seeking as a threat also avoided help-seeking in our class. One interesting finding is that threats and avoidance are positively correlated to instrumental help-seeking instead of negatively correlated as found in previous studies (e.g. Huang & Law, 2022; Pajares et al., 2004; Ryan & Pintrich, 1997). This difference may be due to the nature of instrumental help that students would ask AI for help. In other words, asking AI for help is probably interpreted differently than asking people (either teachers or peers) for help. In addition, our study results found that formal/informal help-seeking is positively correlated with instrumental help-seeking, which is misaligned with the previous research (Karabenick, 2003; Ryan & Pintrich, 1997; Schenke et al., 2015). This misalignment can be due to the definition of the constructs. In traditional help-seeking literature, formal/informal help-seeking asked students to compare their preference between asking instructors (as formal) to asking peers or technology (informal). However, in the current study, we asked the students to compare instructor's/peer's help with AI help. Another possible reason for the misalignment is the context, DGBL. In DGBL, students are likely to switch between seeking help from AI when faced with challenging tasks and seeking help from teachers or peers when they believe they need assistance, indicating students might engage in both types of help-seeking to cover different learning needs. This study was among the first to examine such a relationship in a ChatGPT-embedded DGBL setting, our findings suggest that AI-help and human-help help-seeking behaviors can coexist and contribute to a well-rounded learning experience, allowing students to address challenges, deepen understanding, and satisfy their curiosity within the game-based learning environment.

The second research question examined the relationships among help-seeking profiles and gameplay variables such as highest score, play time, and number of attempts. Consistent with previous literature, instrumental help-seeking is positively significant to students' gameplay such as highest score, game time, and number of attempts (Aleven et al., 2016). Students with instrumental help-seeking engaged in the game more, which led to a better game score. One interesting finding is that students' number of attempts was positively related to avoidance. The more the students avoided help in the game, the more time they played the game. This result can be reasonable if help-seeking was a positive experience in the gameplay. When students were less inclined to avoid seeking help, they may have explored various methods to address challenges in the game, leading them to attempt more frequently (Newman & Schwager, 1995).

The next research question addressed the relationship among students' help-seeking profile and their AI-assisted help-seeking behaviors. Our results showed no significant relationships between students' help-seeking profiles and their AI-help behaviors. This finding aligns with (Huang & Law, 2022) which found that students' participation in online learning was not directly impacted by their perceived help-seeking threat, avoidance, nor instrumental help-seeking. Indeed, Finney et al. (2018) study found that students' help-seeking attitudes and behaviors vary considerably across subjects, ages, and contexts. This study expands the literature on help-seeking profiles and behaviors in relation to ChatGPT and implies students view AI as a potential helping source but do not fully embrace or utilize it. This mismatch can be attributed to various factors such as lack of training or understanding of ChatGPT and valued human interaction over technology. While certain gamers may find it easy to seek assistance during gameplay, other students may perceive asking for help from generative AI as a potential form of cheating. The contextual environment of games and classes may confuse some students regarding the boundary of help-seeking, which is a concern when we implement AI-assisted learning environments. Consequently, it is important for instructors to champion the adoption of generative AI, not only for advancing technological proficiency but also for fostering a positive perception of generative AI among students.

The last research question examined the relationship among students' help-seeking profiles and the learning outcomes. This result is somewhat consistent with the literature (Fong et al., 2023). In the meta-analysis, Fong and his colleagues (2023) found that students' help-seeking profiles, such as avoidant, instrumental, and threats are related to students' outcomes with a very small average weight correlation. Specifically, avoidant and threat are negatively correlated to learning outcomes, and instrumental help-seeking is negatively related to learning outcomes. However, maybe due to the limited sample size, our results were not significant, as the correlation was relatively small ($-0.07 > r > 0.10$). The correlations between threat and learning outcomes, as well as between instrumental and learning outcomes, both fall within the 95% confidence intervals reported in Fong et al.'s study. However, the correlation between avoidance and learning outcomes was weaker compared to the 95% confidence intervals reported in Fong et al.'s study ($r = -0.07$). It is possible that the effect of help-seeking may not be as strong (as we see the correlation between AI-behaviors and posttest was not significant). Another possibility of the non-significant result is the AI-context. The tendency of threats and avoidance help-seeking in an in-person or online classroom would be stronger than help-seeking towards a computer (which is not as well-studied). Students could feel a lot more comfortable asking AI for help. Consequently, the effect of threats and avoidance help-seeking on learning outcomes can be weaker.

Additionally, an incidental finding emerged from this study: a negative correlation was observed between the highest game score and the number of GPT questions asked by players. While this relationship was not a focus of the primary research questions, the observed negative correlation between the highest game score and the number of GPT questions asked provides insight into how players interact with external scaffolding during gameplay. Players who achieved higher scores appeared to rely less on GPT, suggesting that they may have stronger problem-solving skills or more familiarity with the game mechanics. Conversely, players who asked more GPT questions may have been seeking support to overcome challenges, potentially slowing their progression. Although this finding is tangential to the primary focus of the study, it highlights the need for further research to explore how players balance reliance on external scaffolding with independent problem-solving.

While numerous studies have investigated help-seeking in the context of interpersonal interactions, such as with teachers and peers, there has been limited exploration concerning help-seeking in relation to technology (e.g., Mahasneh, et al, 2012; Ryan & Pintrich, 1997). The current study advances our understanding of help-seeking from ChatGPT, a state-of-the-art generative AI tool, as an instrument for help-seeking. Building on past research to understand help-seeking through the lens of self-regulation, the current study aims to discern how help-seeking profiles correlate with educational game behaviors and outcomes. Furthermore, we aim to evaluate the efficacy of ChatGPT in relation to help-seeking profiles, as well as its impact on game behaviors and subsequent learning outcomes.

Implications and limitations

Several practical implications for using ChatGPT in DGBL can be derived from this study. While students interpret asking AI for help differently from asking humans, it is suggested that incorporating AI assistance alongside human assistance can provide students with a variety of resources of support. Game designers can design the DGBL environment in a way that encourages students to utilize both AI and human resources for assistance, and this can enhance gameplay outcomes and motivate students to seek assistance when needed. Instructors should also communicate with students to mitigate stigma and fear of seeking help, both from AI and humans.

Ethical considerations regarding the use of student data and their interaction with AI systems can be a concern in the design and development of educational games. The first issue is students' data privacy as students may not fully comprehend the terms of conditions regarding data privacy when they use generative AI (Kurni et al., 2023). Therefore, educational game designers should pay attention to the task design to minimize the possibility of using personal data in generative AI. Since the task we employed in our game focusing on physics, the risk of sensitive sharing data with ChatGPT was extremely low. Another ethical consideration is student agency and autonomy (Kurni et al., 2023; Moore et al., 2024). To ensure students were fully aware of the experiment that they participated in, all students' participation was voluntary, and they were informed that they could cease participation in the study at any point, with no impact on their grades. In addition, the use of ChatGPT was entirely optional, even within the treatment group. The focus should be on enhancing their learning experience rather than relinquishing control.

There are limitations of the current study that should be noted. First, while the ChatGPT AI-help was intended for help-seeking, the presence of peers and instructors could potentially exert undue influence on students' game performance and use of ChatGPT. Second, future studies might incorporate students' goal orientations, self-efficacy levels, and data related to AI-help access. This could lead to a more holistic understanding of the dynamics of help-seeking and its potential impact. Third, future research should investigate the impact of diverse help-seeking profiles on behavior across various sources, so that we can enhance our comprehensive comprehension of the dynamics involved. Fourth, future studies could build upon our work by including a rigorous evaluation of ChatGPT's reliability and its direct effect on learning outcomes. This would provide a more comprehensive understanding of the tool's role in facilitating learning and help-seeking behaviors in DGBL environments. Fifth, the lack of experimental control is another limitation of the current study. It is suggested that future studies should consider including a control group to provide a comparative analysis of learning outcomes with and without ChatGPT assistance. Such research would be instrumental in comprehensively understanding the impact of AI chatbots on learning effectiveness. Lastly, future studies should include a larger and more diverse group of participants, including students with different prior knowledge, for a more robust generalizability of results.

Appendix: Help-seeking profiles 12-item questionnaire

Subscales	Items	Mean	Std Dev	Min	Max	α
Help-seeking threats	I would feel like a failure if I needed help in playing the game	2.55	1.29	1	5	0.82
	I would not want anyone to find out that I needed help in playing the game	2.20	1.22	1	5	
	Getting help in playing the game would be an admission that I am just not smart enough to do the work on my own	2.34	12.9	1	5	
Help-seeking avoidance	If I didn't understand something in playing the game, I would guess rather than ask someone for assistance	2.48	1.27	1	5	0.72
	Even if the work was too hard to do on my own, I wouldn't ask for help with this class	2.33	1.28	1	5	
	I would rather do worse on an assignment I couldn't finish than ask for help	2.10	1.16	1	5	
Instrumental help-seeking	When playing the game, I would ask AI to explain it to me, not just give me the answer	3.32	1.37	1	5	0.88
	When playing the game, I would ask AI to give me examples of similar problems	3.30	1.31	1	5	
	When playing the game, I would ask AI to give me hints or clues rather than the answer	3.25	1.29	1	5	
	When playing the game, I would ask AI for help so that I could keep working on it	3.41	1.23	1	5	
Formal/informal help-seeking	If I were to seek help in the game, I would ask the teacher or students rather than AI	2.93	12.0	1	5	0.65
	In the game, the teacher or students would be better to get help from than would AI	2.94	1.15	1	5	

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Data Availability The data supporting the findings of this study are available from the corresponding author upon reasonable request.

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